Compute Cannot Buy Identifiability: Information-Limited Two-Attempt Selection in ARC-AGI-2 Solver Pools

An information bound, a conditional compute-separation theorem, and measured evidence from real solver pools and six scored submissions

ARC Prize 2026 Paper Track team

Preprint — not peer reviewed · 18 October 2026

Working draft v6 — 2026-11-01 (verifier module). NOT FROZEN. Derived from frozen v5 (sha256 `a5d9728c…`); v5 remains untouched. v6 keeps all v5 sections, adds **Theorem 8** (verifier-induced identifiability, impossibility certificates, depth allocation, hybrid correction, §3.10), §4.5 verifier experiments, §5.4 large-scale numerical verification, and AWM [42]. Recorded v6 word cap **8,200**. Do not publish until Lead freezes. **No priority claim.**

Abstract

ARC-AGI-2 gives each test input exactly two attempts, scored by exact match. Because distinct candidate grids are mutually exclusive, this two-slot decision is fundamentally a label-free identification problem: the two best candidates can always be submitted, but no metadata rule can choose better than the posterior unless the metadata reveal the truth identity. We prove **Theorem 6**, a label-free identifiability bound: for any (possibly randomized) rule using metadata Z, E[U] ≤ p1+p2+min{T, sqrt(I(C;Z)/2)} (nats) ≤ p1+p2+min{T, sqrt(I_bits(C;Z)·ln2/2)} (bits), where p1,p2 are the top marginals, T the oracle tail and I(C;Z) the mutual information. It is capped by oracle coverage, attained at I=0, vacuous when $I \geq 2T^2$, with a pilot certificate plus an explicit hidden-transfer assumption. On our pools the metadata term is **non-binding** (I=.059–.183 vs $2T^2$=.008–.041), so the certificate is **vacuous**: T6 is a proven, assumption-free bound, not validated by these pools. Unconditional Pinsker bounds are therefore **vacuous in score-dominated pools**; splitting `Z=(S,Z')` gives the score-conditional certificate `E[U] ≤ U_S+min{g−U_S,√(I(C;Z'|S)/2)}`, **tight (equals U_S) under score sufficiency** (Prop. 6f), while Cor. 6g records the negative result that full-metadata Pinsker bounds cannot certify label-free selection. Substantively, on dev20 S2/S5/S5_ens the measured selection gain equals the tail T exactly (ratio 1.00, all generational headroom recovered); on CompressARC the certificate is .200 while the rules **lose −2.25 pp**, consistent with score redundancy. **Theorem 7** (conditional on A1–A3) separates identification from compute: lim_{C⟶∞} score = Σ_t(g_t∞−Δ_t∞) is allocation-independent, so verifier-free compute cannot shrink the gap ΣΔ_t; the hidden-score ladder is consistent (S5 16.25 > S6 = S5_ens 15.42 > S5+LegC 14.17) though confounded by truncation and n=6; a candidate-prefix replay is definitional and not used as T7 evidence. A bounded-DSL verifier accepted **0/120** public-eval tasks (impossibility certificates at depth 2); an n=12 train control certifies the engine on one reachable task, so verifier-induced identifiability is real but coverage-limited (Theorem 8). Six scored Kaggle submissions place our champion at **16.25%** public LB and the same pipeline at **20.0%** on a 20-task public-eval subset, far below the ~30 public-notebook band. No state-of-the-art or medal claim is made.

---

1. Introduction

ARC-AGI-2 measures skill-acquisition efficiency on unseen grid transformations [1, 2]. Its Kaggle incarnation is efficiency-constrained [12]: the notebook re-runs offline for ~12 h without internet, and each input receives exactly two guesses. Because exact matches decide tasks, **selection** is a first-class

decision problem separate from generation: a solver with the right grid in its pool still scores zero if its two attempts miss it.

Public practice ranks candidates by inverse consensus or a likelihood variant [3, 4, 8, 9] and submits the top two distinct grids. We re-derive selection from the scoring rule. Three observations drive the paper: what matters is P(either guess correct), not P(first); correlation is structural because candidates share TTT state, views, beam prefix and model; and a better rule needs calibrated probabilities and correlations.

We contribute a formalization, an information bound, a pre-registered test, real submissions and measured evidence. (i) We define the coverage objective; exact-match distinct grids are mutually exclusive, so the pair problem reduces to label-free identification. We prove **Theorem 6**, a metadata-information bound on achievable two-attempt coverage, with a computable pilot certificate (§3.7), and **Theorem 7** (conditional), a compute-separation theorem showing the identification gap is budget-invariant (§3.8). Lemmas give swap regret, fixed-pool monotonicity, and a descriptive generation×selection identity (§3.9); a budget-allocation setting note replaces the earlier Theorem 5. (ii) We freeze a claim/downgrade protocol and benchmark real third-party pools at two scales plus simulated stress tests (§5). (iii) The measured result is bounded negative: the verdict is `DOWNGRADE_TO_ANALYSIS`, the simulation strongly positive (+18.33 pp over argmax, 95% CI [11.67, 25.0]), but on 400 CompressARC tasks top-2 covers 20.00% while coverage-optimal covers 17.75% against 34.25% oracle; the gap is identification-limited, not reranking-reachable. (iv) We report six scored submissions and GPU measurements (§4), data/leakage audit (§6), open-source system (§7), and landscape analysis (§8).

Nearest prior work: public-notebook heuristics include coverage-aware set selection, MDL-divergence two-slot selection and structural attempt-2 distinctness [17, 23], and rank-based top-2 with minority recoveries [39]. A concurrent instrumentation study of a faithful NVARC reproduction — empirical, not a coverage model — reports a comparable pattern (26.1% vs 32.5% perfect selection; a grouped CV reranker no better than the deployed vote rule) [26]; the headroom observation is already documented. A coverage-vs-selection identifiability gap is formalised for single-answer deployment [28], and on ARC-AGI-1 an unbiased pass@k decomposition finds that selection already recovers ≈95% of the candidate ceiling (generation-bound) [27]. The coverage objective is classical [24, 25, 38]; we provide the two-attempt formalization, calibrated/correlated estimator, and score-sufficiency criterion.

Fresh 2026 work sharpens the positioning. The ARC-AGI-2 Technical Report [29] is the closest published TTT+scoring recipe but offers no attempt-2 theory; Multi-Perspective Transformers [30] report **21.7% eval**, the honest external comparison above our 16.25% hidden / 20.0% dev20. SPLIT [31] allocates compute under an *imperfect verifier*; our hidden setting has no verifier on hidden inputs, so its guarantee does not apply. Correct but Unselectable [32] decomposes coverage into selectable candidates (an interface tax), not an information deficit; Li et al. [33] require a non-degenerate reward tail, while exact-match correctness makes the tail degenerate; Zhou [34] studies generation-side pass@k inversion; Zhang and He [35] require labels; the Pothula dataset [36] and CHARM [37] document plateau/scaling context; concurrent allocation work [40,41] targets posterior success and best-arm identification. None subsumes a label-free coverage bound for k=2 exclusive lists, and none claims the conditional budget-invariance separation; we claim no priority.

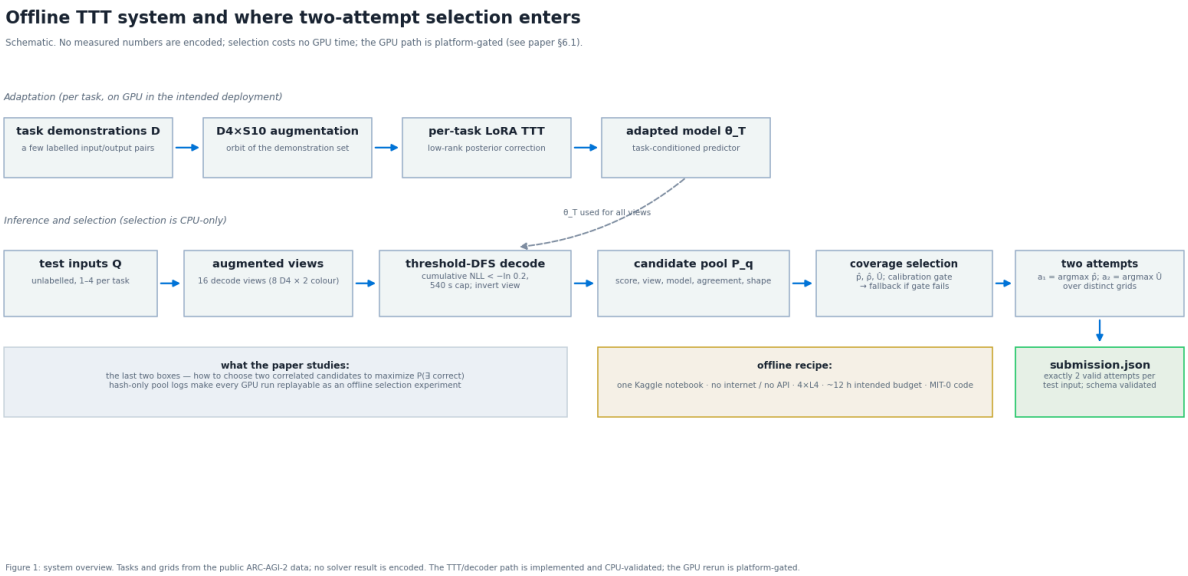


Figure 1. Offline TTT system and where selection enters: demonstrations ⟶ augmented TTT ⟶ threshold-DFS decoding ⟶ candidate pool with scores and view provenance ⟶ calibrated coverage selection ⟶ two-attempt submission. Selection costs no GPU time. Schematic; no measured numbers encoded.

2. Problem formalization and notation

Task and submission. A task is T=(D,Q) with labelled demonstrations D and unlabelled test inputs Q=(q_1,…,q_m); grids are integer arrays over colours 0–9, at most 30×30. For each test input the submission provides an ordered pair of distinct grids (a_1,a_2); C(a)∈{0,1} is the latent event that a is exactly required. The two metric definitions — per-input coverage and task-level all-or-nothing — are both monotone in P(C(a_1)∨C(a_2)); the per-input score averages it, the task-level score is its m-fold product [3, 21], so the objective below is valid under either.

Coverage objective. For attempts a_1,a_2,

$$U(a_1,a_2) := 1 - P(\neg C(a_1) \wedge \neg C(a_2)). \quad (1)$$

Maximizing U differs from maximizing P(C(a_1)): the joint failure term couples the attempts.

Candidate pools from TTT. TTT adapts a pretrained ARC-SFT model θ on augmented demonstrations and decodes under transformed views; for view v (augmentation g_v, beam path, model member) the predicted grid is inverted as $\hat{y}_v = g_v^{-1}(f_\theta(g_v \cdot q))$. The pool for test input q is

$$P_q = \{(y_i, s_i, view_i, model_i, agree_i, shape_i)\}, \quad (2)$$

with s_i a plausibility score, agree_i the view count, and shape_i a demonstration-derived flag. The pool's oracle ceiling bounds any rule's per-input accuracy.

Relation classes. Pairs are grouped by shared failure factors: `same_grid` (correlation 1, never worth two attempts), `same_view` (same TTT state/augmentation), `same_model` (different views), and `cross_model`/`cross_source` (lowest correlation). The rule needs per-class error correlations, observable in metadata without labels.

id	source	benchmark	scale / unit	published anchor	role
B	`vanductan/arc-agi-2-candidate-dump` (displayed LB 30.97)	ARC-AGI-2 public eval	20 tasks / 29 test inputs; unit = input; 347 candidate files	oracle-any-2 9/29 = 31.0%; source top-2 8/29 = 27.6%	primary, **pilot** (CIs mandatory)
A	CompressARC [6] bundled histories (Liao & Gu 2025; MIT)	ARC-AGI-1 train/eval	400 eval tasks (fit 400 train); unit = task; hash-level	pass@2 80/400 = 20.00%; any-guess 137/400 = 34.25%	secondary, large-N replication
C	Parameterised simulated pools	synthetic	many seeds; known ground truth	oracle by construction	theory stress test + estimator sanity

Table 1. Evaluation datasets (frozen).

3. Theory

3.1 Independent case: top two calibrated probabilities

Theorem 1. If correctness indicators are independent with p_i=P(C_i), then P(¬C_i∧¬C_j)=(1−p_i)(1−p_j), so

U(i,j) = p_i + p_j − p_i p_j, (3)

strictly increasing in each argument. The coverage-optimal pair is therefore the two largest p_i, and attempt order is irrelevant. This is the only regime where first-guess ranking is coverage-optimal — the assumption behind inverse-consensus top-2.

3.2 Duplicates are strictly dominated

If a_2 is the same grid as a_1, C(a_2)=C(a_1) and U(a_1,a_2)=P(C(a_1))=p_{a_1}; any distinct candidate with p>0 strictly increases coverage. **Theorem 2.** Submitting the same grid twice is strictly dominated by any distinct candidate with positive correctness probability.

3.3 Correlation breaks argmax-once

Candidates share TTT state, augmentation view, beam prefix, and model. For two Bernoulli correctness indicators with correlation coefficient ρ,

U(i,j) = p_i + p_j − p_i p_j − ρ√(p_i(1−p_i)p_j(1−p_j)), (4)

from P(¬C_i∧¬C_j)=(1−p_i)(1−p_j)+Cov(C_i,C_j). Two same-view candidates with perfectly correlated failures (ρ=1) force equal marginals p_i=p_j=p: their pair coverage is U=p. An independent third candidate with p_3>0 gives U=p+p_3−p p_3>p; at p=0.50, p_3=0.35 the gain is +17.5 coverage points on that input. **Theorem 3.** Under positive error correlation, the pair maximizing the first guess is not in general coverage-optimal; a decorrelated candidate can strictly improve coverage. For ρ<1 the pairwise preference condition between the runner-up j and a decorrelated candidate k is

(1−p_1)(p_k−p_j) > ρ_{1k}√(p_1(1−p_1)p_k(1−p_k)) − ρ_{1j}√(p_1(1−p_1)p_j(1−p_j)). (5)

Diversification pays when the correlation penalty exceeds the probability advantage — typical for per-view errors; zero or negative correlation does not force it, which is why $\hat{\rho}$ is measured rather than assumed.

Corollary. If the ranking score is cluster mass and views are independent, top-2 by mass coincides with top-2 by posterior probability, so the rule reduces to inverse consensus — a strict generalization that departs from it under correlation.

3.4 Diversification threshold (Theorem 4)

Theorem 4 (exact, finite-pool). Let two distinct candidates have correctness probabilities `p_1 ≥ p_2` (both in [0,1]) and correctness-indicator correlation `ρ = Corr(C_1,C_2)`. Then

U(pair) = p_1 + p_2 − p_1p_2 − ρ√(p_1(1−p_1)p_2(1−p_2)), (6)

and the pair strictly beats the top-1 **duplicated** attempt (`U_dup = p_1`) iff

ρ < ρ(p_1,p_2) := √(p_2(1−p_1) / (p_1(1−p_2))). (7)*

Proof. `U(pair) − U_dup = p_2(1−p_1) − ρ√(p_1(1−p_1)p_2(1−p_2))`; its sign is positive iff `ρ² p_1(1−p_1)p_2(1−p_2) < p_2²(1−p_1)²`, and, for interior `p`, dividing by the positive `p_1p_2(1−p_1)(1−p_2)` gives `ρ² < p_2(1−p_1)/(p_1(1−p_2))`; both sides are non-negative for interior `p`, so

taking square roots gives (7). ∎ Equivalently, for interior p_1,p_2, ΔU := U(pair) − U_dup = √(p_1(1−p_1)p_2(1−p_2))·(ρ* − ρ) — the gain is the distance below the threshold scaled by the candidates' Bernoulli spread. ρ* ∈ [0,1] for interior p (ρ*=0 at boundary p_2=0 or p_1=1, where ΔU=0), with ρ* = 1 iff p_1 = p_2; negative correlation always clears the threshold; ρ = ρ* is exact indifference.

Corollary 1 (equal marginals). If p_1 = p_2 = p then ρ* = 1 and ΔU = p(1−p)(1−ρ): diversification pays for every ρ<1, maximized at ρ=0 and worthless at p∈{0,1}.

Corollary 2 (skew shrinks the threshold). For fixed ρ>0, diversification pays iff p_2 > ρ²p_1/(1−p_1+ρ²p_1); for small probabilities ρ* ≈ √(p_2/p_1), so any fixed positive correlation is eventually dominated by skew.

Corollary 3 (score sufficiency and exact-match exclusivity; corrected 2026-10-26). For non-exclusive verifiers, a score tie cannot distinguish candidates, and if correctness is score-determined (Proposition 5) tied candidates share correctness (ρ=1, forcing p_1=p_2), so ΔU=0. For **exact-match** ARC, distinct candidate grids are mutually exclusive: C_i∧C_j=∅, hence ρ<0 always, the Theorem-4 gate is always met, and U(a_1,a_2)=p_{a_1}+p_{a_2} for any joint distribution; the optimal pair is the two largest calibrated p_i (Corollary 4). The real-pool null is therefore estimation/identifiability, not positive correlation: oracle headroom exists, but p̂ cannot separate oracle-gap candidates from exposed metadata (Proposition 5).

Derived gate and falsifier. For non-exclusive verifiers, deploy diversification across a relation class only when ρ̂c < ρ*(p̂_1,p̂_2) − δ (δ=0.05, frozen); otherwise take the duplicate/consensus alternative — the gate is derived, not tuned. The empirical instantiation is falsifiable on held-out units: **P1** admitted pairs beat their top-1 duplicate (paired Δ>0, 80% CI>0); **P2** rejected pairs do not (Δ≤0); **P3** the predicted-vs-realized gain slope lies in [0.7,1.3]. Failure falsifies the estimator/relation-class model, never the algebra; protocol and tool are in Appendix E. The threshold test ran on dev20 pools, but exact-match mutual exclusivity makes its positive-ρ premise inapplicable: every class has ρ̂<0 (same-model/cross-view −0.13, score-tied −0.18), so no deployable pair is rejected and P1/P2/P3 are untestable (§4.8). This is an EXPLORATORY check, not evidence; Table 10 is a prose check and Fig. 9 is omitted.

Table 10 (replaced). Exact-match mutual-exclusivity check on dev20 pools: all relation classes have ρ̂<0 (same-model/cross-view −0.13, score-tied −0.18), P2′s rejected set is empty; EXPLORATORY, no P1/P2/P3 claim (theory §4.8).

3.5 Setting observation: budget allocation (formerly Theorem 5)

Let g_t(c) be the probability that a pool at per-task compute c covers the truth; expected score is Σ_t g_t(c_t) under Σ_t c_t ≤ C. **Theorem 5.** (i) Identical concave g_t ⟹ uniform c_t = C/T is optimal (Jensen). (ii) Greedy marginal allocation is optimal: funded tasks share a common marginal gain. (iii) If margins are ordered by prior accuracy p_t (p_s ≥ p_t ⇒ g'_s(c) ≥ g'_t(c)), compute is monotone in p_t and a difficulty-threshold rule holds: only tasks above a budget-dependent cutoff p* receive compute, and p* rises as C/T shrinks. (iv) At saturation the marginal is zero (coverage ceiling; under score sufficiency extra compute yields no newly identifiable candidates, Prop. 5), so excess is better reallocated; never-solvable tasks receive none. This is a **setting observation**, not a new theorem: SPLIT [31] derives verifier-aware allocation, and [40,41] give fixed-budget posterior/best-arm allocation bounds; T6/T7 below carry the paper's new claims.

3.6 Calibrated plug-in estimator

Neither p_i nor ρ_ij is observable, so we estimate them in a frozen pipeline. (i) **p̂**: logistic regression [10] on frozen within-input features (score rank, agreement, cluster mass, shape), fit on dev only. (ii) **ρ̂**: per relation class from the dev co-failure statistic, clipped to [−0.95,0.99], with frozen pooling and a default for sparse classes. (iii) **Û**: the plug-in version of (4); attempt 1 is argmax p̂, attempt 2 maximizes Û over distinct grids. (iv) **Gate/fallback**: deploy only if dev cross-fit calibration has Brier ≤0.10, reliability slope in the code's valid range [0.2,5.0] (design target [0.7,1.3]) and an estimable relation class; otherwise use

a no-calibration fallback (highest agreement, then a different grid). The claim is pre-registered at ≥+1.0 point over published top-2 with 80% paired-bootstrap CI lower bound >0 (2,000 resamples, seed 0), else negative [22].



Figure 4. Illustrative schematic of the coverage rule: candidate pool grouped by relation class, calibrated p̂/ρ̂, pair coverage U, and the chosen distinct pair. Toy values only — not measured.

3.7 Identifiability-limited two-attempt coverage (Theorem 6)

Exact-match scoring makes distinct candidate grids mutually exclusive: `C_i∧C_j=∅` for `i≠j`, so for any pair `A`, `U(A)=Σ_{i∈A}p_i` with no independence or correlation assumption. The prior-optimal pair is `{1,2}`, `U0=p1+p2`, achievable by ignoring metadata; `g=Σ_i p_i` is oracle coverage and `T=g−p1−p2` the tail mass. Whether any label-free rule can beat `U0` is therefore an information question, not a ranking question.

Theorem 6 (label-free identifiability bound). For every (possibly randomized) rule using metadata `Z`, E[U] ≤ E_Z[top-2 posterior mass] ≤ p1+p2+min{T, sqrt(I(C;Z)/2)} (nats) ≤ p1+p2+min{T, sqrt(I_bits(C;Z)·ln2/2)} (bits),

capped by the oracle `g`. *Proof:* top-2 posterior mass is 1-Lipschitz in total variation, Pinsker gives `TV≤sqrt(KL/2)`, Jensen gives `E[TV]≤sqrt(I/2)`, and the tail cap is `E[top2]≤g`. ∎ The bound is attained at `I=0` by the constant rule, equals `g` when `Z` identifies `C`, and is **vacuous** when `I≥2T²` (the tail cap binds). The gain is at most `T`; Pinsker is order-tight, not an equality.

Proposition 6e (non-vacuity). The unconditional T6 bound is strictly informative iff `I(C;Z)<2T²` nats; otherwise it equals the oracle `g`. Every pool here satisfies `I≥2T²`, so the unconditional certificate is **vacuous** — an honest caveat, not validation. **Corollary 6g (negative result about the method).** In score-dominated exact-match pools, Pinsker-type bounds on full metadata **cannot certify label-free selection**: the bound collapses to the oracle cap. **Proposition 6f (constructive fix).** Split `Z=(S,Z')`, where `S` is score-only information and `Z'` residual metadata: `E[U] ≤ U_S + min{g−U_S, sqrt(I(C;Z'|S)/2)}`, with `U_S` the best score-only rule's coverage; the bound is informative iff `I(C;Z'|S) < 2(g−U_S)²`, and if score sufficiency (Prop. 5) holds then `I(·|S)=0` and the bound equals `U_S` exactly. This is the paper's sharpest new theoretical statement: score dominates, and only residual metadata can help.

Computable pilot certificate. With a labeled pilot, estimate `p̂1,p̂2,p̂3`, `T̂=ĝ−p̂1−p̂2`, and `Î(C;Z)` by binning exposed metadata (score margins, rank gaps, agreement, cluster mass, shape, model count) against truth identity; bootstrap task-level CIs. For the evaluated population, `gain ≤ min{T̂, sqrt(Î/2)} + O(sqrt(log(1/δ)/n))`. Hidden tasks have **no labels**, so Î is not estimable there: this is a pilot certificate plus an explicit exchangeability transfer assumption, never a hidden-task guarantee; with partial metadata, `I` is a lower bound and `B` may overstate the achievable bound.

dataset	n	p̂1/p̂2/p̂3	H(C)	Î (feature)	B [95% CI]	measured gain	cert.
CompressARC	400	.185/.015/.020	.547	.071 (view)	.200 [.160,.238]	**−2.25 pp**	vacuous
pilot16	16	.313/0/0	0	.183 (view)	.313 [.125,.563]	0 pp	vacuous
dev20 S1	32	.063/.125/0	.637	.167 (margin)	.188	0 pp	vacuous
dev20 S2	32	.094/.063/.063	1.079	.071	.156	+6.25 pp	vacuous
dev20 S5	32	.125/.031/.063	.956	.059	.156	+6.25 pp	vacuous
dev20 S5_ens	74	.162/.054/.014	.753	.059 (vote)	.216 [.122,.311]	**+8.11 pp**	vacuous

Table 12. D1 identifiability certificates on real pools (uncalibrated consistency proxies; certificate vacuous on every row because $I \geq 2T^2$).

The +8.11 pp S5_ens figure is selectable headroom not realized by our submitted aggregator (S5_ens hidden 15.42 < S5 16.25). Because $I \geq 2T^2$ the certificate equals the oracle cap and is **vacuous**: these pools do not test T6. The substantive observation is the gain/T ratio: dev20 S2/S5/S5_ens recover T exactly (1.00), while CompressARC loses −2.25 pp against a .200 certificate (score redundancy, Prop. 5). Falsifiers: F1 held-out gain outside the certificate CI; F2 large Î with zero realized gain; F3 vacuous bound with zero gain; F4 monotonicity violation.

Prop. 6f addendum (measured). Table 12b evaluates the score-conditional certificate with `S` = score-only strata. It is **non-vacuous on exactly one dataset, CompressARC**: U_S=.200, g−U_S=.1425, I(C;Z'|S)=.0336 nats, cap'=.1296 (12.96 pp), and the measured rule loss (−2.25 pp) lies inside it. Elsewhere the bound remains vacuous (0 violations). The score-sufficiency prediction that I(C;Z'|S)=0 and the bound equals U_S is **not testable here**: I(C;Z'|S) is .083–.225 nats, not ≈0, and pilot/dev20 score strata collapse to two bins (sparse, flagged), so we can neither confirm nor reject tightness. This tests the certificate's non-vacuity, not that metadata helps.

dataset	U_S	g−U_S	I(C;Z'\	S) nats	cap'	measured gain	informative
CompressARC	.200	.1425	.0336	**.1296**	−2.25 pp	**TRUE**	
pilot16	.3125	0	.2255	0	0 pp	FALSE	
dev20 S1	.1875	0	.1334	0	0 pp	FALSE	
dev20 S2	.15625	.0625	.0829	.0625	+6.25 pp	FALSE	
dev20 S5	.15625	.0625	.0921	.0625	+6.25 pp	FALSE	
dev20 S5_ens	.216216	.081081	.0903	.081081	+8.11 pp	FALSE	

Table 12b. Prop. 6f score-conditional certificates (`S` = score-only strata; informative iff `I<2(g−U_S)²`; 0 violations).

Check of non-vacuity, not evidence metadata helps; the one non-vacuous row still loses to score-only selection.

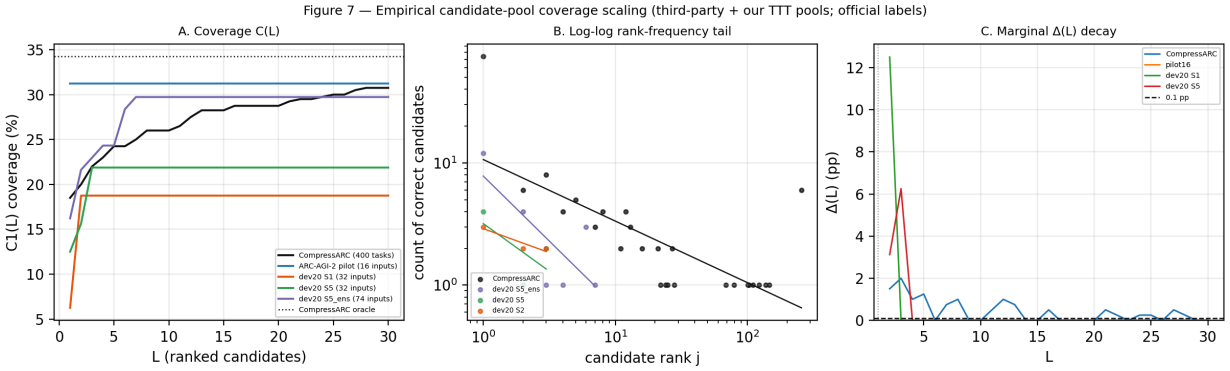


Figure 7. Prefix coverage versus candidate-prefix size L under the rank power-law model: C2-prefix at L=2/5/10/20 for CompressARC (20.0/24.3/26.0/28.8, $\hat{\alpha}$=1.83 [1.67,2.06], $L^* \approx 257$), pilot16 (31.3 flat), dev20 S1 (18.8 flat), S5 (15.6/21.9/21.9/21.9, $\hat{\alpha}$=3.42, $L^* \approx 3$) and S5_ens (21.6/24.3/29.7/29.7, $\hat{\alpha}$=2.75 [2.13,4.60], $L^* \approx 7$). Fixed-solver top-2 coverage is reported separately in Table 8.

3.8 Compute cannot substitute for identifiability (Theorem 7; conditional)

Assumptions. (A1) `g_t(c)` is nondecreasing and saturating in compute, with limit `g_t∞`. (A2) The identification gap `Δ_t(c)=g_t(c)−u_t(c)` (oracle minus best label-free coverage) satisfies `Δ_t(c)≥Δ_t∞>0`

for unidentifiable tasks and does not vanish with compute; refinement buys rank candidates, not identification. (A3) generation and verification are separable at the relevant scale.

Theorem 7 (conditional separation). Under A1–A3, `lim_{C→∞} score = Σ_t(g_t∞–Δ_t∞)`, independent of the allocation; the verifier-assisted ceiling is `Σ_t g_t∞`; the gap `Σ_tΔ_t` is budget-invariant; greedy allocation is optimal only on the identifiable part. *Interpretation:* verifier-free compute cannot buy identifiability; what grows with budget is candidate coverage, which is only useful where metadata can identify the candidate. This explains why our S6, S5_ens and S5+LegC submits did not beat S5.

Empirical support. The defensible support is the **hidden-score ladder**: S5 16.25 > S6 15.42 = S5_ens 15.42 > S5+LegC 14.17 — more components/compute did not help. This is **consistent, not conclusive**: n=6, truncation at the 578 s cap, S6's selection bug, S5_ens's 53-task superset, packet/config differences and unknown hidden size confound it. A candidate-prefix replay (Table 13) is **definitional** — top-L coverage for units whose truth rank exceeds L rises trivially — so we do not use it as T7 evidence.

dataset	unidentifiable n / slope	identifiable n / slope
CompressARC	343 / 0.0 [0,0]	57 / **19.1** [15.9,22.1]
pilot16	16 / 0.0 [0,0]	0 / —
dev20 S2/S5/S6	30 / 0.0 [0,0]	2 / **20.0** [20.0,20.0]
dev20 S5_ens	68 / 0.0 [0,0]	6 / **26.7** [23.3,30.0]

Table 13. D2 candidate-prefix replay (definitional; not T7 evidence).

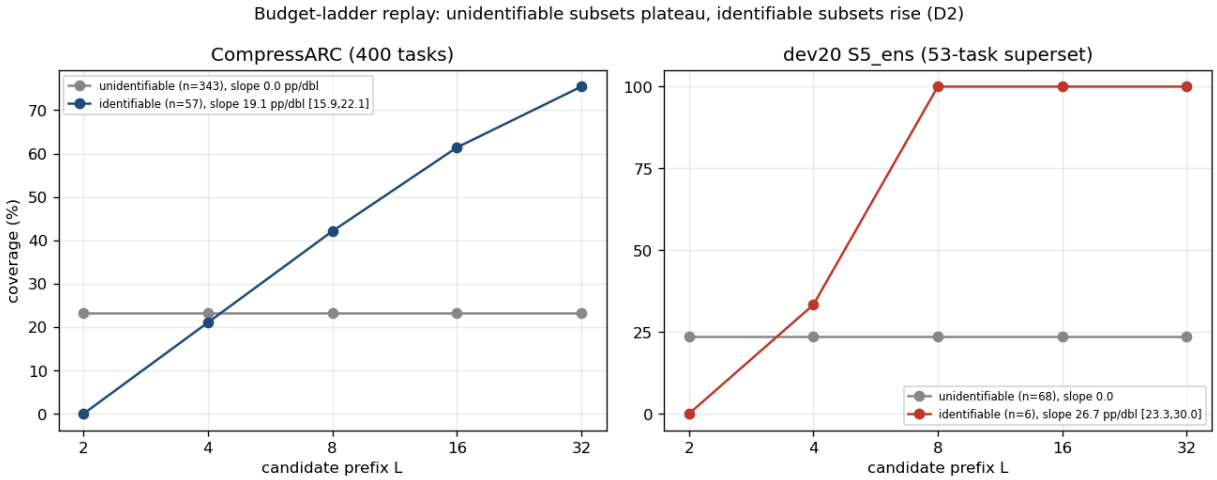


Figure 8. Candidate-prefix replay (D2): unidentifiable subsets plateau across `L`, identifiable subsets rise trivially by construction. Definitional; not evidence for T7.

3.9 Supporting lemmas

Lemma 6a (swap regret). If the top-2 pair `{1,2}` is replaced by `{1,3}`, coverage drops by exactly `p2−p3` under exclusivity; `p2−p3` is therefore the *cheapest local-swap regret*, not a subtractive gain term in any upper bound. **Lemma 6b (fixed-pool monotonicity).** For a fixed candidate pool, re-ranking cannot reduce two-attempt coverage; it is strict iff a new distinct candidate with positive posterior enters the top-2. Consequently pass@k inversion cannot arise from selection alone: it requires generation or rank-distribution drift, consistent with the generation-side diagnosis of Zhou [34]. **Lemma 6d (generation×selection identity).** Pool coverage factors as `g(L)·s(L)`, where `g` is oracle coverage and `s` the selection ratio; this identity is descriptive only and makes no scaling-law claim.

3.10 Verifier-induced identifiability (Theorem 8)

Selection can be bypassed when executable hypotheses are verifiable against the demonstrations. Let `H_d` be the programs reachable at depth `d`, `A_d={h∈H_d: V_D(h)=1}` the accepted set (train-exact fit), and partition `A_d` into behavior classes `j` (programs with identical test predictions); `q_j` is the class posterior under a stated prior, `g_V(d)=P(truth∈A_d)`, with assumptions **A0** (stated prior over accepted programs), **A1** (train-fit soundness only), **A2** (reachability: a reachable truth program is accepted and

has its own class). AWM [42] instead explores executable Python hypotheses, verifies them against observations and formalizes them in Lean 4 (reachability/impossibility proofs); we adapt only the verifier step: demonstrations are labels, and verification restores identifiability. CHARM [37] reports 46.7% ARC-AGI-2 public-eval pass@2 as external SOTA context only; we do not reuse its code (licence unavailable). Distinct classes predict distinct grids, so correctness events are exclusive.

(i) Unique class ⇒ gap closed. If A_d has one behavior class, then conditional on truth∈A_d verifier-only coverage is 1, unconditionally g_V(d), and the identifiability gap is **0**; if truth∉A_d, coverage is 0. *Proof:* every accepted program fits D; with one class and a reachable truth, all accepted programs predict the truth grid, and exact-match exclusivity makes the event vacuous otherwise. ∎

(ii) General case. The best verifier-only two-attempt rule submits the top-2 class grids, so C_V(d)=g_V(d)·Σ_{j≤2}q_j and the residual gap is Δ_V(d)=g_V(d)(1−Σ_{j≤2}q_j); with fewer than two classes the second attempt must come from TTT or a backup. *Proof:* conditional on truth∈A_d the truth class is distributed as q; two attempts cover iff it is among the top-2; multiply by g_V(d); exclusivity gives the sum, not the product. ∎

(iii) Impossibility certificates and depth allocation. A_d=∅ gives verifier coverage **0 at depth d**: a certificate of failure at that depth, not of unsolvability. With c_t(d)=g_{V,t}(d)s_t(d) nondecreasing in depth and per-depth cost λ, min Σλd_t s.t. Σ_t c_t(d_t)≥K has the water-filling condition Δ_t(d_t)=μ for funded tasks and ≤μ for unfunded ones; under single-crossing it is a threshold rule, and tasks with identically zero marginal acceptance are skipped. *Proof:* separable monotone allocation by the Theorem-5 exchange argument with depth in place of compute. ∎

(iv) Comparison with T7. Theorem 7′s verifier-assisted ceiling Σ_t g_t is attainable only if g_{V,t}=g_t with unique class (or top-2 class mass 1); otherwise the verifier-assisted ceiling is Σ_t g_{V,t} s_t<Σ_t g_t. The cost is certifying search, not labels: labels are free (train pairs), coverage is not.

Hybrid correction. The product form 1−(1−p1)(1−p2) is incoherent for distinct grids; exact-match gives P=p1+p2.

Frozen notation (T8 hybrid). Let U=1 denote exactly one verifier behavior class at depth d*, with unique grid v and precision φ=P(v=truth|U=1) (φ=1 under A2). Let w be the TTT top-1 grid and w2 the TTT runner-up; p_T=P(truth=w), p_w2=P(truth=w2) are TTT marginals (working assumption: TTT marginals independent of U, testable). Verifier-first submits (v,w) when U=1 and w≠v; when w=v it falls back to w2; otherwise (w,w2). Coverage is the exclusivity sum E[1{truth=v}+1{truth=w,w≠v}+1{truth=w2,w=v}] (no product term); with multiple classes the top class replaces v at posterior q1. Verifier-first beats TTT-first iff E[1{U}(φ−p_w2)]>0; if verifier coverage is zero (U=0 a.s.) it reduces exactly to p_T+p_w2.

Scope: conditional on A0–A2; a weak DSL makes g_V small and the theorem vacuous; test-time soundness is measured, not assumed. Falsifiers: F1a re-verification (0 violations = PASS); F1b unique-class precision (PASS if φ̂≥0.9, 80% lower ≥0.8, n_unique≥20); F2 hybrid complementarity; F3 acceptance growth by depth.

3.11 TTT as amortized Bayesian adaptation

Why are TTT pools plausible? Read ARC-SFT training as approximate posterior inference [13]: the parameters are a prior over task rules, and per-task TTT performs amortized Bayesian adaptation. A LoRA update is a low-rank posterior correction; a PAC-Bayes sketch predicts an interior optimum in rank/steps and larger gains where the prior is weakest. Test-time augmentation is an equivariance prior, valid only where the task rule commutes with the transform; the frozen κ_T proxy records this. It is a sketch, not a bound.

4. Accuracy: the offline TTT system and a real public submission

4.1 First submission attempt (errored)

We entered [15] with one MIT-0 notebook (Qwen3-4B ARC-SFT, D4×S10 augmentation, per-task LoRA TTT, threshold-DFS decoding, NLL rescoring, coverage selection, schema validation). Submission 57456856 (kernel v8; packet `3774c892…`; S1/aug8) used the 41,400 s budget and 578 s/240-task cap (1156 s/120); hidden size unknown. Kaggle reports COMPLETE at ≈02:35 UTC after ≈2 h 34 min, **no publicScore**, errorDescription *"Your notebook hit an unhandled error while rerunning your code…"*. Cause: rerun branch `ARC_BUDGET_S="41400.0"` parsed by `int("41400.0")` → `ValueError` before try/except. Hardened packet adds int-typed/defensive parses, no-raise validation, a SAFETY cell; probe 57500870 scored 0.00. **We claim no score for 57456856 and report the failure as engineering, not model quality.**

Final hidden-test results (six scored submissions): **S5 16.25** (ref **57681632**, kernel v16; champion), **S6 15.42** (57716141), **S5_ens 15.42** (57752011), **S1 14.17** (57663405), **S5+LegC 14.17** (57785740), CPU probe **0.00** (57500870); plus one invalid-format attempt (57442788) and one rerun error (57456856, Table 9). No score exceeds 16.25. A reference public-notebook replication attempt reached **3/20** on dev20, below our S1 4/20, so the ~30 public-notebook band is external context, not a matched reproduction. The LegC verified-induction hybrid was **net-negative** on the hidden half (−2.08 vs S5) and was rejected. Compute quota was exhausted Oct 31–Nov 1, leaving no GPU for tuning; the Nov 2 final entry uses the best completed version (S5 v16) unless S5b beats 16.25. Half-cost configs truncate at the 578 s cap: these scores are far below the ~30 band, not SOTA.

57456856 exposes no per-task diagnostic; labeled diagnostics come from dev pools (§4.3) and third-party pools (§5).

date (UTC)	ref	config	LB	status / notes
2026-10-20T14:56	57442788	CPU-fallback insurance	—	invalid format
2026-10-21T00:01:15	57456856	S1 v8, packet `3774c892…`	—	rerun-only `int("41400.0")` ValueError; no publicScore
2026-10-22T00:00:23	57500870	S1 CPU-version probe	**0.00**	COMPLETE, no error
2026-10-26T10:45	57663405	S1 v8 L4, packet `9060264c…`	**14.17**	COMPLETE, no error (≈14 h incl. queue)
2026-10-27T00:00:28	57681632	**S5 v16 L4**, packet `6211b694…`	**16.25**	COMPLETE, no error — **champion**
2026-10-28T00:00:27	57716141	S6 dual-config v22	**15.42**	COMPLETE
2026-10-29T00:02:23	57752011	S5_ens two-model v23	**15.42**	COMPLETE
2026-10-30T00:02:21	57785740	S5+LegC hybrid v25	**14.17**	COMPLETE — LegC net-negative
2026-10-31	—	(submission missed)	—	session gap; no ref
2026-11-01T00:10:23	57854697	S5b v24, packet pending	pending	PENDING at freeze — last S5-family probe

Table 9. Submission ledger (all public-LB attempts; errors and misses included).

Hidden-set size remains unknown; runtimes are queue time, not a size signal. Six scored entries; all scores are public-LB hidden-half results. Table 8′s dev20 scores are public-eval-subset measurements, not hidden scores.

4.2 Hidden-test budget calibration

Hidden size is unknown (120–240); for the 240/259 sample shape at 11.5 h, the four-worker budget is 41,400 s and the fair per-task cap is 578 s at 240 tasks (1156 s if 120; 9.6/19.3 min), including a reserve. The cap truncates attempts. Dev2 timings: ≈220 s/task (202.51/238.12 s). Scaling to 240 tasks on four workers gives ≈3.7 h of TTT — 2.6× per-task headroom, ~7.8 h unused wall. Slack absorbs harder tasks, model load (106.9 s) and worker skew. Processed-task count and truncation are not observable

from the rerun interface. Figure 9 reserves this calibration and the §4.3 dev compute-accuracy points. Theorem 5 predicts compute beyond a coverage plateau is wasted; the dev20 ladder tests this (Table 11; EXPLORATORY/PENDING).

4.3 Compute-normalized dev evidence

On this single 20-task subset (no CI claim), the all-config union (incl. S5_ens) reaches 6/20 (30%) versus 4/20 for S1; four-config S1/S2/S5/S6 union 5/20. Dev2 pilot: 466.7 s wall, 106.9 s load, 15 candidates for 2/2 inputs, oracle 1/2 — plumbing evidence, not accuracy.

config	dev20 (task)	dev20 (input)	final LB	notes
S1 (hardened)	4/20 = 20.0%	6/32 = 18.8%	**14.17**	decode mean ≈235 s/task; 4 tasks capped
S2 (aug16)	2/20 = 10.0%	5/32 = 15.6%	—	decode up to 618 s; truncation
S5 (single)	3/20 = 15.0%	5/32 = 15.6%	**16.25**	champion
S6 (v13; selection-bug)	2/20 = 10.0%	5/32 = 15.6%	**15.42**	pass@2 flagged; 4/20 oracle, usable
all-config union	6/20 = 30.0%	—	—	cand.-union incl. S5_ens; four-config 5/20

Table 8. dev20 sweep (eval20: 20 tasks / 32 inputs; L4×4; hardened packets).

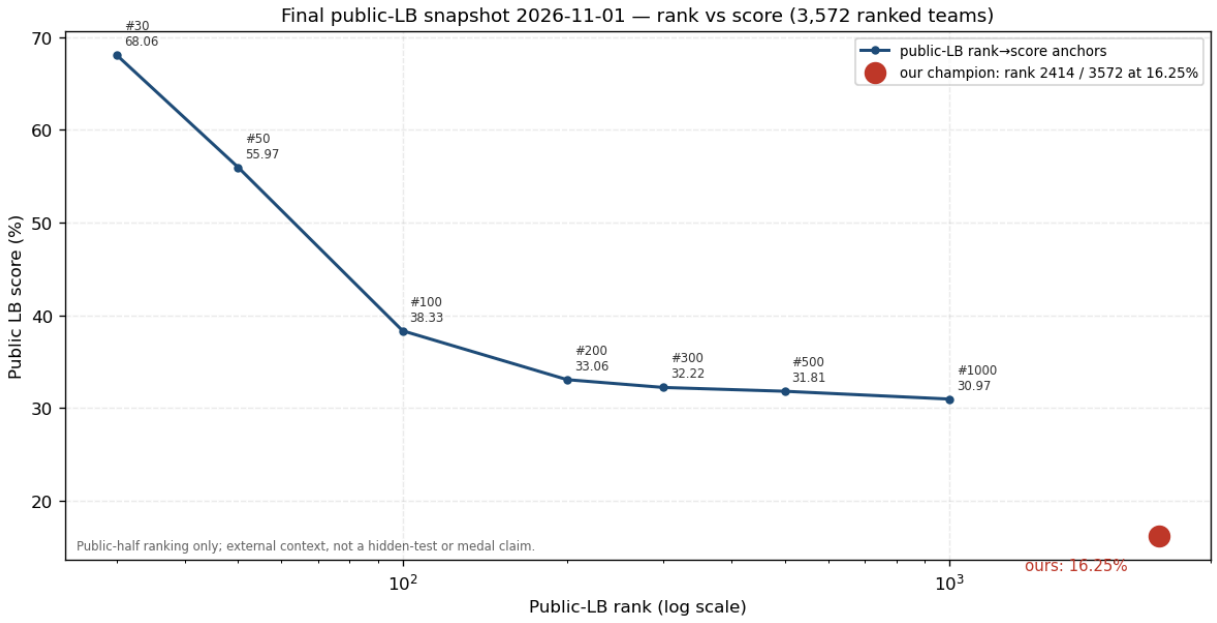


Figure 9. Final public-LB snapshot (2026-11-01T00:13Z): rank↔score anchors and our champion at rank 2414 / 3,572 (score 16.25). Public-half ranking only; external context, not a hidden-test or medal claim.

Table 11. Public-LB snapshot 2026-11-01T00:13:02Z. 3,572 ranked teams; leader 88.06. Our row: `rank 2414 · "wen Chen" · TeamId 17156552 · score 16.25 · submissions 8 · last 2026-10-30 00:02:21` (three teams tied at 16.25, ranks 2412–2414; 67.5th percentile from top). Anchors: r30 68.06, r50 55.97, r100 38.33, r200 33.06, r500 31.81, r1000 30.97. The public LB displays our **best** score (16.25), not the last submission (v25 = 14.17): standings use best-of-submissions semantics, so the Nov 2 entry is only a safety copy for the last-submission rule. The ~30–31 notebook band is ≈ranks 500–1000 (top 14–28%); 16.25 is far below it — public-half ranking only; no medal/SOTA claim.

4.5 Verifier experiments: bounded-DSL results

We tested an AWM-style verifier-first search (LegC DSL; T0 identity, T1 atoms, T2 composition) on public eval120. The depth-2 engine accepted **0/120** programs: unique coverage 0%, attempt-1 precision undefined, **certificate fraction 120/120 impossibility**, 334.7 CPU-min total (2.79 CPU-min/task; 56 workers). F1a is PASS but vacuous (nothing accepted to re-verify); F1b is INCONCLUSIVE (no unique classes); F2′s pre-registered verdict is **FAIL** (hybrid = TTT top-2; on the exploratory 20-task subset S1

4/20 = 20.0% vs verifier-only 0%, H–T=0, 80% CI [0,0]); because verifier coverage is zero, the complementarity hypothesis is untestable in interpretation and the hybrid reduces exactly to TTT top-1+runner-up, consistent with T8′s coverage premise failing on AGI-2. F3 **FAILS at measured depth 2** (acceptance flat for all 120; depth 3 not measured and not imputed). Learned-rule families (0/120), DeepSeek-chat (0/20), and DeepSeek-reasoner (0/10; all completions token-truncated) give nulls; the reasoner null is **infrastructure-limited, not capability evidence**. The n=12 training control yields **1/12 unique solve (0520fde7)**, confirming the engine can certify reachable programs; it is small and not used for a rate claim.

run	n	accepted	unique	empty-certificate	note
eval120	120	0	0	120/120	334.7 CPU-min, 56 workers
train control	12	1	1	11/12	unique 0520fde7; 1/12, no rate claim
learned rules	120	0	0	—	0.3 min
LLM chat	20	0	0	—	DeepSeek-chat k=4
LLM reasoner	10	0	0	—	token-cap truncation: infrastructure null

Table 14. Verifier search runs (public eval120; labels used for scoring only; not a competition entry).

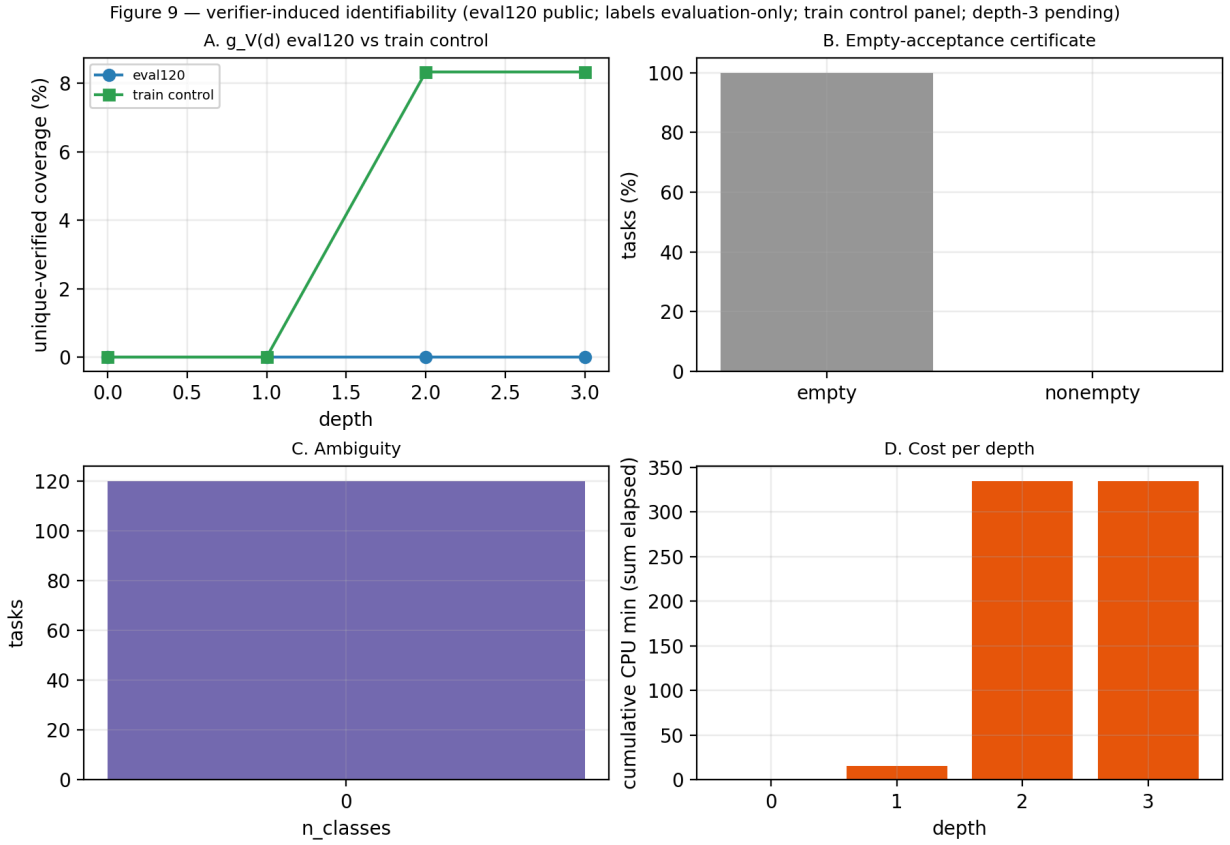


Figure 9. Verifier search on public eval: (A) `g_V(d)` for eval120 vs the n=12 training control; (B) impossibility-certificate fraction by depth; (C) class ambiguity; (D) cost per depth. Public eval, labels scoring-only, not a competition entry; depth 3 not imputed.

5. Empirical study: selection rules on third-party candidate pools

5.1 Sources, reproduction gates, and protocol

Released third-party logs bound the study: gains are pool-conditioned; sources are frozen and a failing gate excludes one. Table 3 gates: B's coverage summary (29 inputs; oracle 9/29; top-2 8/29) and A's artifact (137/400 solved; 80/400 pass@2) reproduce exactly. Rules: duplicate, argmax, top2, cal_indep,

coverage_optimal, oracle_any2 (ceiling), deployed. Splits are task-level (B 9/11 tasks, 13/16 inputs; A 400/400); paired bootstrap over tasks (2,000 resamples, seed 0) [14] gives 80%/95% CIs; no refits.

source	gate	published	our reproduction	pass
B	coverage summary	29 inputs; oracle 9/29; top-2 8/29	exact, 0 per-input mismatches	yes
A	upstream script, 2000 steps	solved 137/400; ≤2 guesses 80/400	exact via bundled artifact	yes

Table 3. Reproduction gates (source-published metric vs our re-derivation).

5.2 Pilot evidence on Dataset B

B is the only public candidate-dump notebook; 20 tasks, pilot-only, holdout 16 inputs/11 tasks. Oracle 31.25% (5/16); every deployable rule (duplicate, argmax, top-2, cal_indep, coverage) also 31.25%, indistinguishable (80% CI [18.75,43.75]); headroom 0. Deployed fallback 25.00%.

5.3 The large replication and the simulated stress test

A is claim-bearing: top-2 covers 80/400 (20.00%) vs 137/400 any-guess (34.25%). The result is a null: correctness is predictable but score-dominated. Missed truths are far from the top (rank median 12). `DOWNGRADE_TO_ANALYSIS`; deployed fallback 19.25%; CompressARC [6] is ARC-AGI-1-era (hash-level pools).

C isolates the estimator; a sanity check, not evidence: calibrated rules reach the oracle 41.67% (+18.33 pp, 95% CI [11.67, 25.0]; gate slope 0.025).

rule	B: ARC-AGI-2 TTT pools (16 inputs / 11 tasks)	A: CompressARC (400 tasks)	C: simulated (120 tasks)
oracle_any2 (ceiling, not a rule)	31.25 [18.75, 43.75]	34.25 [31.25, 37.50]	41.67 [35.83, 47.50]
duplicate	31.25 [18.75, 43.75]	18.50 [16.00, 21.00]	16.67 [12.50, 20.83]
argmax1_argmax2	31.25 [18.75, 43.75]	19.50 [17.00, 22.00]	23.33 [18.33, 28.33]
top2_consensus	31.25 [18.75, 43.75]	20.00 [17.50, 22.75]	21.67 [16.67, 26.67]
cal_indep	31.25 [18.75, 43.75]	18.00 [15.50, 20.75]	41.67 [35.83, 47.50]
coverage_optimal	31.25 [18.75, 43.75]	17.75 [15.25, 20.25]	41.67 [35.83, 47.50]
deployed (gate-selected)	25.00 [12.50, 37.50]	19.25 [16.75, 21.75]	41.67 [35.83, 47.50]

Table 2. Frozen two-attempt rules on the three datasets (pass@2, %; 80% bootstrap CI in brackets; seed 0, 2,000 resamples; two decimals).

Cells are pass@2 from the frozen artifact `analysis/results/selection_benchmark.json` (sha256 `6c16abfef7f0efe3a1a6bbe9f6251c9aeef1893ee7b0f44fdfdb9c2eb34ab71a`; 80% CIs). C is simulated, not evidence about real solvers.

Score sufficiency explains the null: score ordering saturates recoverable coverage, so selection is not the bottleneck — generation is; re-ranking never exceeds top-2′s 20.00%, the 14.25-point headroom is real but not separable from exposed metadata. Falsifiers: (F1) a calibrated rule beating the solver by ≥+1.0 point (80% CI lower >0); (F2) metadata exposing oracle-gap candidates; (F3) a multi-decoder union exceeding noise.

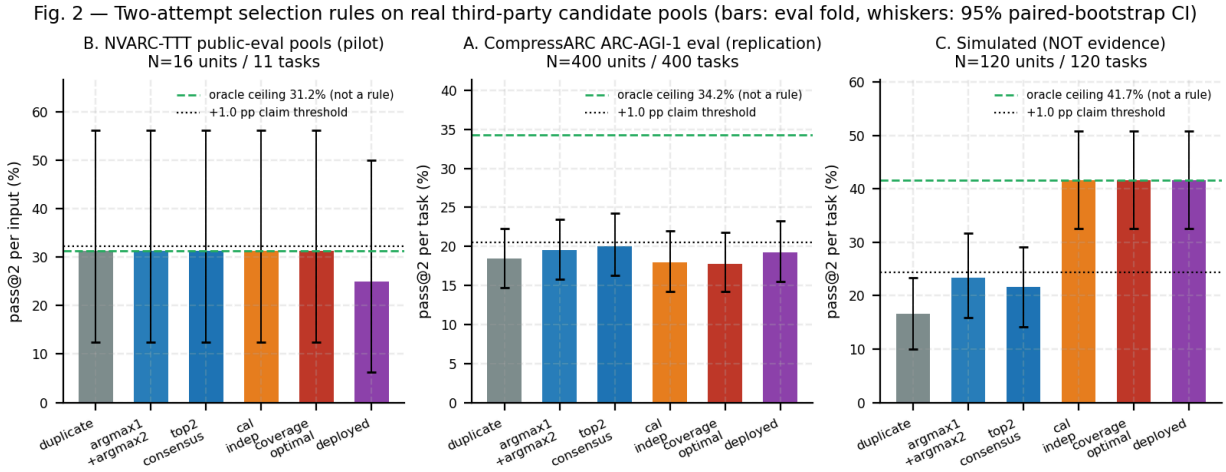


Figure 2. Rule comparison with bootstrapped CIs: pass@2 and paired Δ vs the published baseline, per dataset (B pilot; A replication; C simulated). Oracle coverage is a ceiling, not a rule; bars are the pre-registered 80%/95% intervals; the +1.0-point line applies to the claim-bearing dataset only.

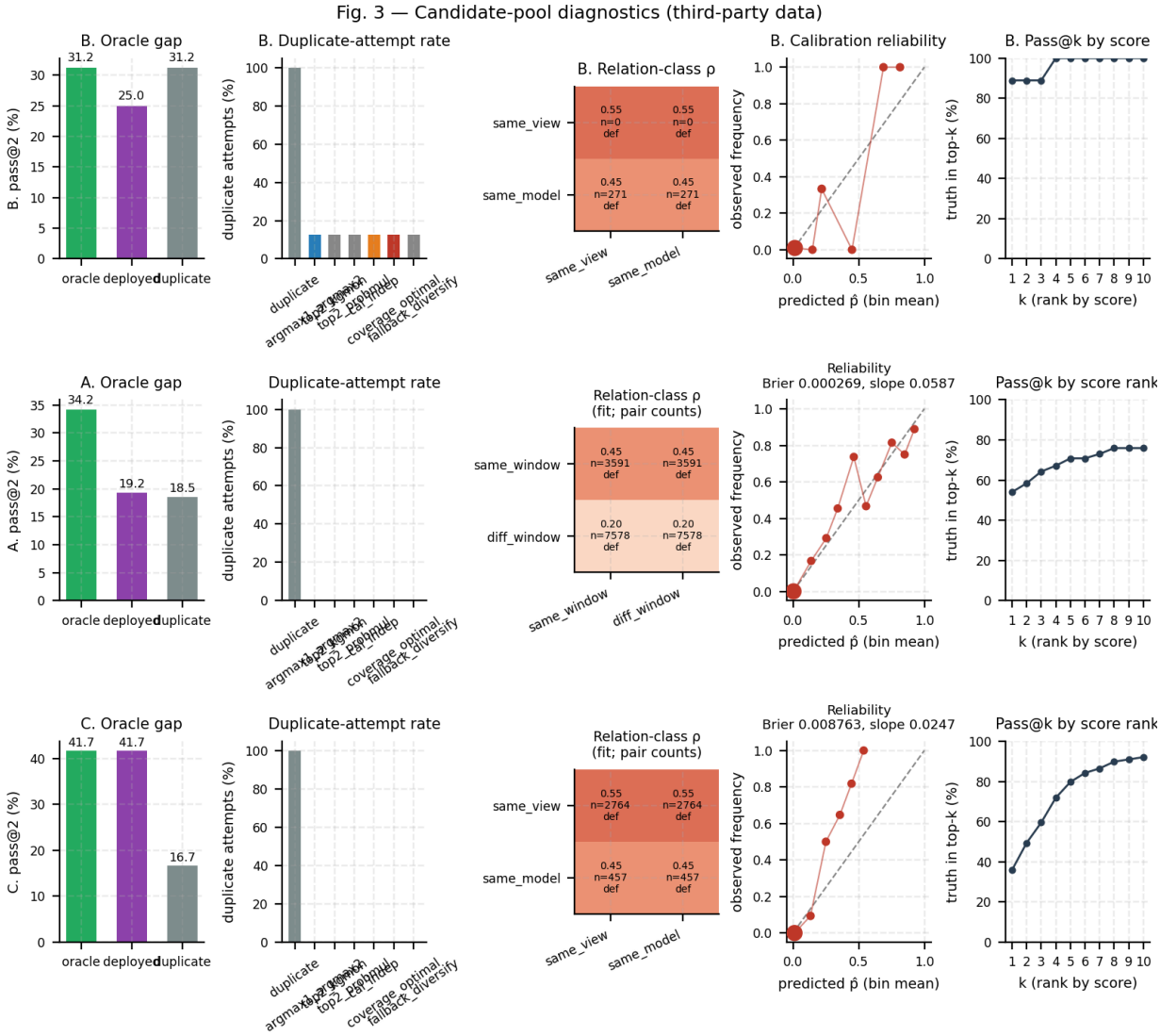


Figure 3. Pool diagnostics: oracle gap, duplicate rates, relation-class ρ heatmap with pair counts, and calibration reliability for B/A and C.

5.4 Large-scale numerical verification of T6/T7

Two cloud-box runs check the theorem implementations at scale. **T6:** 100,000,000 synthetic instances (`bound_sim_gpu_100m.json`) produce 30,063 floating-point excesses, **all below the pre-registered 1e-6 tolerance (zero violations)**; maximum excess `1.12e-7`, mean bound-cap gain .209, 28.3 s. **T7:** 2,000,000 tasks (`alloc_sim_gpu_2m_v2.json`) give mean `δ=.260316`, label-free limit .739684, verifier ceiling 1.0, and gap `.260316 = mean δ` — budget-invariant across the C=50–5000 ladder; water-filling beats uniform by 3.5–8.8× at every rung, 20.1 s. These validate the implementations, not ARC performance. Artifacts: `analysis/results/gpu_sim/bound_sim_gpu_100m.json` sha `b1117f3812ab5200…` and `alloc_sim_gpu_2m_v2.json` sha `e39cd282796366ae…` (superseded alloc v1 `629de32d…`), both from `code/lab/gpu_theory_sim.py` (published in the MIT-0 repo; same file for CPU/GPU). Commands: `python code/lab/gpu_theory_sim.py bound --n 100000000 --out out/bound_sim_gpu_100m.json` (seed 7; K=8 rank categories; acc~U[0.1,0.9]; τ prior p_j∝j^{−α}, α~U[0.5,3.0]) and `python code/lab/gpu_theory_sim.py alloc --n-tasks 2000000 --budget 50 --out out/alloc_sim_gpu_2m_v2.json` (seed 11; τ=exp(N(0,0.8)−0.5) clamped [0.05,20]; identifiable fraction 0.35; δ∈[0.2,0.6] for unidentifiable, 0 otherwise; g(c)=1−exp(−c/τ); bisection 80 iters). Device: NVIDIA TITAN Xp 12,288 MiB, driver 580.105, torch 2.5.1+cu118, CUDA 11.8; 56 vCPU / 94 GB RAM; 9.16 TFLOPS FP32 (4096^3 matmul ×20); runtimes 28.3 s (bound) and 20.1 s (alloc).

check	n	result	time
T6 bound	1e8	0 violations at 1e-6; max excess 1.12e-7	28.3 s
T7 allocation	2e6 tasks	gap = mean δ = .260316; water-filling 3.5–8.8× uniform	20.1 s

Table 15. GPU-scale verification.

6. Data characterization and leakage audit

We characterize ARC-AGI-2 data [20]: 1,000 training tasks (3,232 pairs, 1,076 inputs) and 120 evaluation tasks (358 pairs, 172 inputs); the repo mirror (167 inputs) is not used for scores. A SHA-256 audit finds zero shared tasks, pairs and input grids between training and evaluation.

Heuristic baselines solve 1.6% training, 1.5% holdout and 0% eval; eval grids are larger (21.95 vs 14.67, $p<0.001$) and 54.5%/57.5% of train/eval tasks are same-size rewrites.

quantity	value	artifact
train / eval tasks	1,000 / 120	§1
shared tasks / pairs / input grids	0 / 0 / 0	§12
trivial-heuristic solve: train / holdout / eval	1.6% / 1.5% / 0%	§13
max grid dimension: train vs eval	14.67 vs 21.95 ($p<0.001$)	§11
solvability tiers A/B/C	28 / 56 / 36 (uncalibrated)	§15

Table 6. Data-characterization summary (public ARC-AGI-2; unless noted).

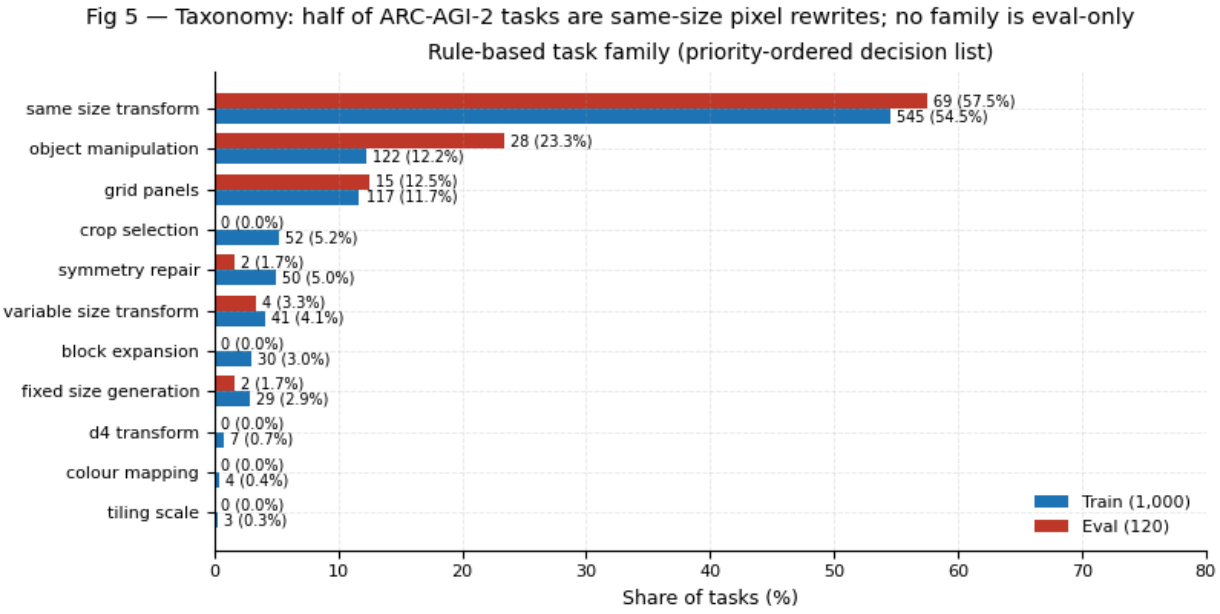


Figure 5a. Rule-based task-family distribution for 1,000 training and 120 public-evaluation tasks (analysis/report.md §9). Same-size rewriting dominates both; no family is evaluation-only.

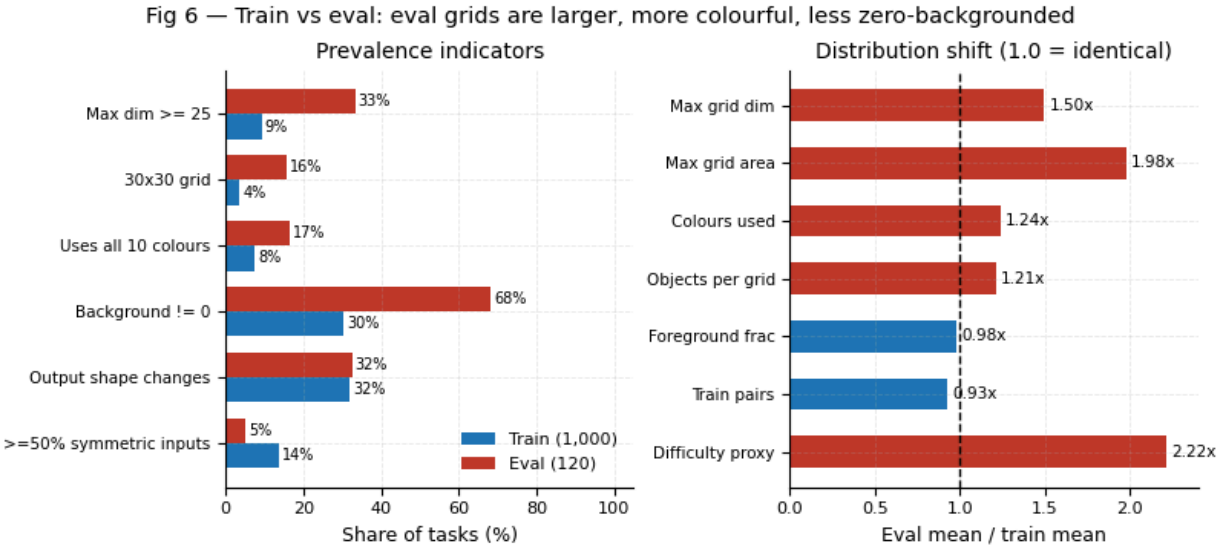


Figure 5b. Train-vs-evaluation shift: prevalence indicators (left) and eval/train mean ratios (right) for size, palette, demonstrations, and background (analysis/report.md §11).

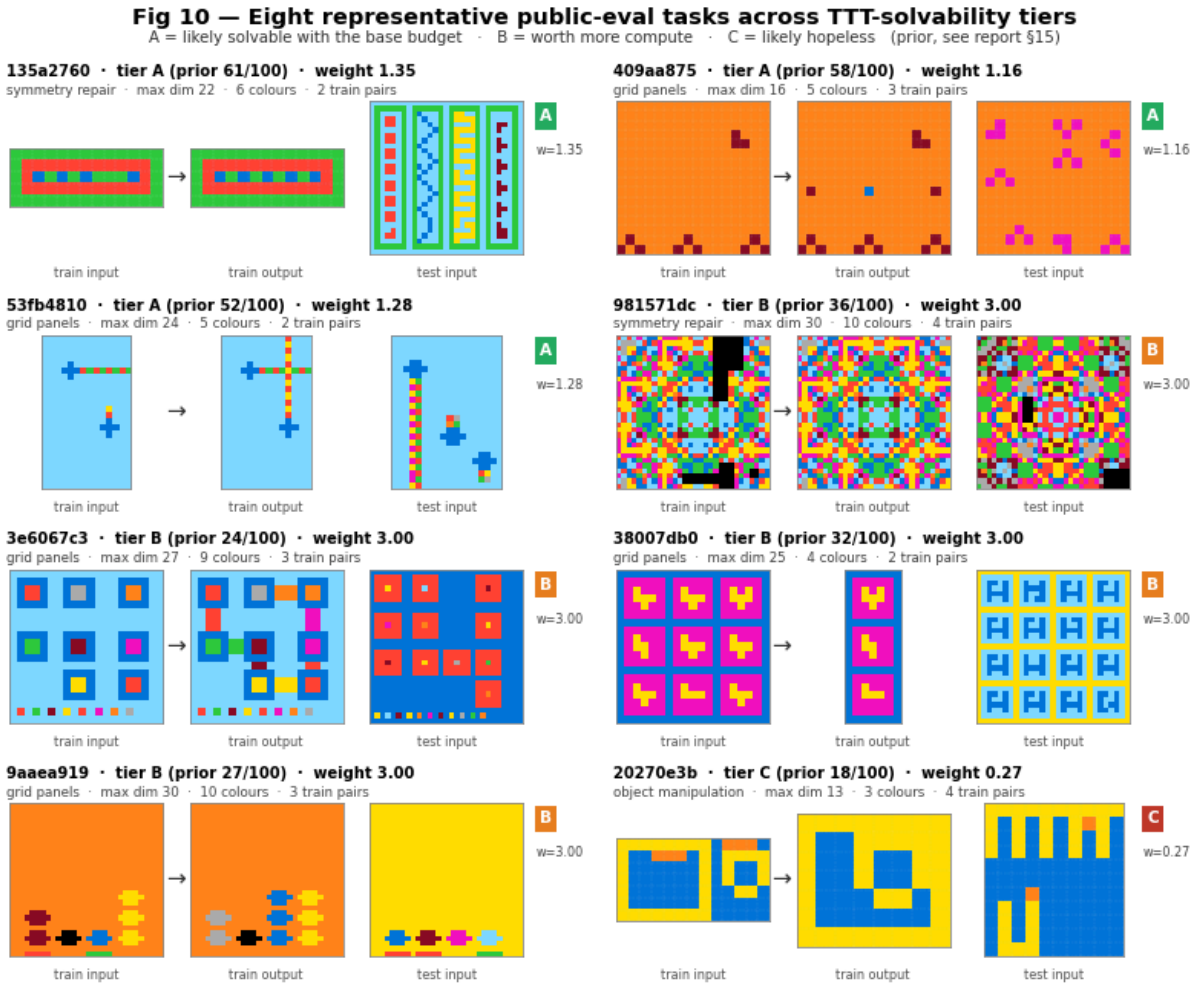


Figure 6a. Eight representative public-evaluation tasks across prior tiers with budget weights (analysis/report.md §15); tiers are an uncalibrated prior, not a solve rate.

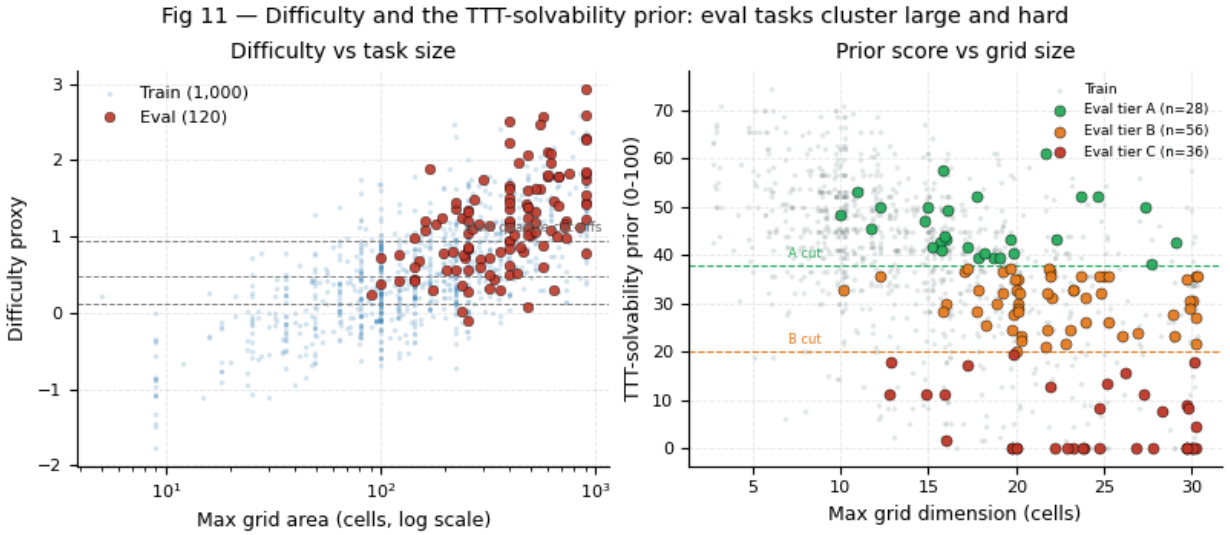


Figure 6b. Difficulty proxy and the uncalibrated TTT-solvability prior versus task size (analysis/report.md §15–16).

7. Completeness: engineering and reproducibility

The repository implements loader, augmentation, threshold-DFS decoding, rescoring, aggregation, coverage rules (§3), TTT and orchestration; it runs offline from a kernel mount. Released under

MIT-0; third-party components retain licences (ARC-AGI-2, Qwen3, CompressARC, arc-dsl, Unsloth/peft). Notebook methods are clean-room re-implemented and cited; a NOTICE records attributions. The `LICENSE` file must switch to SPDX `MIT-0` before release.

GPU fixes: Triton needs an explicit `TRITON_PTXAS_PATH=/usr/local/cuda/bin/ptxas` (missing ptxas → zero candidates), and Hugging Face loading needs offline weights (`local_files_only`/mount). The packet pins source/hash and forces deterministic fallback; Validation: 190 passing CPU tests (~26 s); a schema-valid fallback: the 167-input packet ran in 8.3 s on Kaggle, while the official 172-input repackaging measured 5.4 s locally and scores 0/120 — a safety net, not accuracy; TTT-path validation with a tiny Qwen model (dialect guard, LoRA, 2 steps, 8-view decode, selection, schema); and a CompressARC harness that re-derives bundled predictions and exports hash-level pools for Dataset A.

component	validation evidence	status
CPU test suite	190 tests pass (~26 s)	measured
CPU fallback submission	repo 167-input packet 8.3 s on Kaggle; official 172-input repackaging 5.4 s local, 0/120 (safety net)	measured
TTT path (tiny model)	tokenizer guard, LoRA, decode, selection, schema OK	measured
GPU dev2 pilot	2 tasks, 15 candidates, per-task walls 202.5/238.1 s, load 106.9 s	measured
Engineering fixes	ptxas/HF-offline guards; void stale-packet run root-caused	measured
First submission attempt	57456856 errored (rerun-only budget parse); CPU-probe retry (57500870) publicScore 0.00	measured failure
First scored hidden run	**57663405**, S1 L4, COMPLETE **publicScore 14.17** (≈14 h incl. queue)	measured (scored)
Best scored hidden run	**57681632**, S5 L4, COMPLETE **publicScore 16.25** (current best; +2.08 pp over S1)	measured (scored)
S6 dual-config rerun	Oct 28 submission	PENDING
CompressARC reproduction	137/400 solved; 80/400 pass@2	measured

Table 5. System and reproducibility matrix.

8. Public-solver landscape and what transfers

Table 4 has ten notebooks [16–19]; nine cluster at 29.7–31.4. TRM [5], SOAR [7], deep ensembles [11] are not evaluated.

Saved artifacts are partial reruns (vanductan 5/120; 8/172; others 2–4/120).

notebook (family)	displayed	claimed	mechanism	transferable element
mayankraaz	31.39	35.97	triple-pass pooled TTT, controller, LOO-DSL	cheap-first ordering; pass-A invariant
yuvrajlamba01	31.39	54.31%	TTT + deterministic symbolic solver	monotone symbolic filler
rabari9999	30.97	n/a	v38 perf patch clone	GPU-side log-softmax patch
vanductan	30.97	n/a	perf-patched research dump	candidate-pool logging (Dataset B)
mikelou1 / anthonyxlyu	30.14	33.89	v38 minimal perf patch	algebraically identical speed patch
woominyo	29.72	n/a	perf patch + interleaved ensemble	attempt ordering via interleaving
dmitriigluzdov	29.17	n/a	+6 transformer layers	negative result — capacity expansion
delongyang	15.69	n/a	27B FP8 zero-shot	model-size/throughput trade-off
cprs	30.56	80%+ target	TTT + LLM program-synthesis pre-pass	verified-program candidate source

Table 4. Public-solver landscape (dated reading 2026-10-18; displayed scores are secondary-source context).

9. Limitations, ethics, and reproducibility

The largest limitation is that our solver has **six scored hidden submissions** spanning **14.17–16.25** (champion S5 16.25), versus **20.0% task-level** on the 20-task public-eval subset — S1/S5 gaps ≈5.8/≈3.75 points consistent with half-cost truncation and subset noise, not distributional. Compute quota was exhausted Oct 31–Nov 1, leaving no GPU for further tuning; the ~30 public-notebook band is external context only. The first rerun (57456856) errored after ≈2 h 34 min with no publicScore; probe 57500870 scored 0.00. No per-task hidden observability exists, so variance cannot be reported. The selection claim remains a third-party-pool study with narrow conclusions; B is tiny (29 inputs/20 tasks), A overlaps ARC-AGI-2 training, C validates only estimator/implementation.

D1/D2 caveats. The D1 inputs are empirical consistency proxies, not true posteriors; metadata are partial (third-party dumps; pilot/CompressARC lack shape/view metadata), so `I` is a lower bound and the certificate may overstate achievable coverage; hidden-task transfer is an explicit exchangeability assumption, not a guarantee. Label gaps: `/tmp/dv15/logs` and `logs/pools` were missing, so archived dev20 paths were used; dev20 has only 16/32 units and wide CIs; S6 is selection-bug-affected; S5_ens is a 53-task superset; the +8.11 pp S5_ens headroom is selectable, not realized (hidden 15.42 < S5 16.25). T6 is a **negative result** plus a conditional certificate: certificates are vacuous on all pools ($I \geq 2T^2$); gain=T holds on dev20 S2/S5/S5_ens (ratio 1.00) while CompressARC loses −2.25 pp; Prop. 6f is tight only under score sufficiency. D2 is definitional, not T7 evidence; T7′s only support is the hidden-score ladder, **consistent not conclusive** (n=6; truncation; S6 bug; 53-task superset; packet/config differences; unknown hidden size). **Verifier caveats.** Verifier results are public-eval diagnostics: the bounded-DSL engine accepted 0/120 (impossibility certificates for the bounded class, not all programs); F1a is vacuous, F1b/F3 are inconclusive/fail-at-depth, F2 is untestable with zero verifier coverage; the n=12 train control is small; LLM nulls are token-cap infrastructure-limited; labels are used for scoring only and never inside the search; no hidden-test or SOTA claim. Theory is bounded: coverage identities are exact, but $\hat{p}$ and $\hat{\rho}$ are estimates; the gate is a guard, not guarantee; TTT is a sketch.

All data are public (ARC Prize data, notebook outputs, CompressARC artifacts); no personal data, credentials or hidden-test information is used. Third-party artifacts are used with attribution; no third-party code was copied. Reproducibility depends on Appendix B artifacts and on the repository reaching a single consistent licence before release.

10. Conclusion

Coverage is the right formalization for two-attempt selection: independent correctness makes top-2 optimal, correlated errors make a calibrated pair rule provably better, and Theorem 4 gives the correlation threshold. The measured answer is a bounded negative: on real public pools the solver's score ordering saturates recoverable coverage. On 400 CompressARC tasks top-2 covers 20.00% while the coverage rule covers 17.75%, against 34.25% oracle — 14.25 points exposed metadata cannot reach. The verdict is `DOWNGRADE_TO_ANALYSIS`; selection is not the bottleneck, candidate generation/coverage is. The information view sharpens the result: Theorem 6 bounds any metadata rule by `p1+p2+min{T,sqrt(I/2)}`, so the 34.25% vs 20.00% gap is an identification deficit, not a ranking failure; the certificates are vacuous on every pool ($I \geq 2T^2$), so T6 is proven rather than empirically separated; on dev20 S2/S5/S5_ens selection recovers the tail T exactly (ratio 1.00), while CompressARC loses −2.25 pp, and the S5_ens headroom is selectable but unrealized. Theorem 7 (conditional) makes the separation explicit: verifier-free compute leaves `ΣΔ_t` unchanged while identifiable coverage scales; the hidden-score ladder (16.25 > 15.42 = 15.42 > 14.17) is consistent but confounded, not conclusive. Theorem 8 adds the verifier route: unique accepted classes restore identifiability, impossibility certificates bound search depth, and the bounded-DSL engine certifies 0/120 eval tasks (train control 1/12), so verification cannot close the gap without reachability. Transferables: the formalization, information bound, protocol, benchmark, data audit, implementation, real attempt and record — plus falsification targets: held-out gain outside the certificate, large $\hat{I}$ with zero gain, a vacuous bound with zero gain, or a monotonicity violation.

References

1. Chollet, F. *On the Measure of Intelligence.* arXiv:1911.01547, 2019. https://arxiv.org/abs/1911.01547
2. ARC Prize. *ARC Prize 2026 competition and paper-track rules.* https://arcprize.org/competitions/2026 (accessed 2026-10-18).
3. Sorokin, I., Puget, J.-F. *NVARC solution to ARC-AGI-2 2025* (Kaggle 1st place; model `sorokin/qwen3_4b_grids15_sft139`, Apache-2.0), 2025. https://www.kaggle.com/models/sorokin/qwen3_4b_grids15_sft139 ; https://github.com/1ytic/NVARC
4. Koushik Rudra. *failed-in-aimo* public Kaggle notebook (NVARC-style TTT reference), 2026. https://www.kaggle.com/code/koushikrudra/failed-in-aimo
5. Jolicoeur-Martineau, A. *Less is More: Recursive Reasoning with Tiny Networks (TRM).* arXiv:2510.04871, 2025. https://arxiv.org/abs/2510.04871
6. Liao, I., Gu, A. *ARC-AGI Without Pretraining (CompressARC).* 2025. https://iliao2345.github.io/blog_posts/arc_agi_without_pretraining/ ; code (MIT): https://github.com/iliao2345/CompressARC
7. Pourcel, J., Colas, C., Oudeyer, P.-Y. *Self-Improving Language Models for Evolutionary Program Synthesis: A Case Study on ARC-AGI (SOAR).* arXiv:2507.14172, 2025. https://arxiv.org/abs/2507.14172
8. the ARChitects. *ARC-AGI-2 2025 solution writeup.* 2025. https://lambdalabsml.github.io/ARC2025_Solution_by_the_ARChitects/
9. Cole, J., Osman, M. *Don't throw the baby out with the bathwater: How and why deep learning for ARC.* arXiv:2506.14276, 2025. https://arxiv.org/abs/2506.14276
10. Guo, C., Pleiss, G., Sun, Y., Weinberger, K. *On Calibration of Modern Neural Networks.* ICML 2017. https://arxiv.org/abs/1706.04599
11. Lakshminarayanan, B., Pritzel, A., Blundell, C. *Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles.* NeurIPS 2017. https://arxiv.org/abs/1612.01474
12. Snell, C., et al. *Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters.* arXiv:2408.03314, 2024. https://arxiv.org/abs/2408.03314
13. Akyürek, E., et al. *The Surprising Effectiveness of Test-Time Training for Abstract Reasoning.* arXiv:2411.07279, 2024. https://arxiv.org/abs/2411.07279

14. Efron, B., Tibshirani, R. *An Introduction to the Bootstrap.* Chapman & Hall, 1993. DOI: 10.1201/9780429246593
15. Kaggle. *ARC Prize 2026 ARC-AGI-2 code-competition metadata* (kernels-only, no-internet rerun). https://www.kaggle.com/competitions/arc-prize-2026-arc-agi-2 (accessed 2026-10-18).
16. vanductan. *arc-agi-2-candidate-dump* public Kaggle notebook (displayed LB 30.97; candidate-pool research dump), 2026. https://www.kaggle.com/code/vanductan/arc-agi-2-candidate-dump
17. mayankraaz. *arc-agi-2-triple-pass-nvarc-sota-35-9-lb* public Kaggle notebook, 2026. https://www.kaggle.com/code/mayankraaz/arc-agi-2-triple-pass-nvarc-sota-35-9-lb
18. woominyo. *arc26-nvarc-lb33-4xl4* public Kaggle notebook, 2026. https://www.kaggle.com/code/woominyo/arc26-nvarc-lb33-4xl4
19. mikelou1. *arc-agi2-lb33-89-minimal-perfpatch* public Kaggle notebook, 2026. https://www.kaggle.com/code/mikelou1/arc-agi2-lb33-89-minimal-perfpatch
20. ARC Prize. *ARC-AGI-2 public data release* (Apache-2.0). https://github.com/arcprize/ARC-AGI-2 (accessed 2026-10-18).
21. ARC Prize. *ARC-AGI-2 2026 scoring rules and leaderboard snapshot* (secondary rendering, 2026-10-18). https://arcprize.org/competitions/2026/arc-agi-2
22. ARC-2026 team. *Selection protocol v2: pre-registered evaluation on third-party candidate pools* (internal pre-registration, frozen 2026-10-18; released with the paper's artifact bundle).
23. D. Zhong. *arc-agi2-nvarc-plus-ckpt* (v6s) public Kaggle notebook, 2026 (uses "coverage-aware set selection" and a P4 structural-distinctness attempt-2 rule; the referenced theory notes v32/v33 are not public). https://www.kaggle.com/code/dalezhong/arc-agi2-nvarc-plus-ckpt
24. Wei et al. *FlexRouter: Coverage-Oriented Correlated Subset Selection* (COLM 2026). https://arxiv.org/abs/2609.38585
25. Singham. *Selection Under an Imperfect Verifier: An Extreme-Value Theory of Test-Time Compute.* Zenodo DOI 10.5281/zenodo.21976219, 2026. https://doi.org/10.5281/zenodo.21976219
26. M. N. Jacob. *arc-budget: where test-time compute goes in an ARC-AGI-2 system* (software; empirical instrumentation study, not a formal coverage model), 2026. MIT-0. https://github.com/manunicholasjacob/arc-budget — associated writeup "Most of the Budget Buys Nothing, and a Fifth of What It Finds Is Thrown Away".
27. K. Moghe and P. Chin. *Cost-Effective Agent Harnesses for Abstract Reasoning and Generalization on ARC-AGI-1.* arXiv:2607.06764 [cs.AI], 2026 (preprint; no venue). https://arxiv.org/abs/2607.06764
28. Bay and Yearick. *When More Sampling Hurts: The Modal Ceiling and Correlation Ceiling of Test-Time Scaling.* arXiv:2606.28661, 2026 (coverage-vs-selection ceilings, "identifiability gap"). https://arxiv.org/abs/2606.28661
29. W. L. de Oliveira, M. Bobokhonov, M. Caorsi, A. Podestà, G. Beltramo, L. Crosato, M. Bonotto, F. Cecchetto, H. Espic, D. T. Salajan, S. Taga, L. Pana, J. Carthy. *ARC-AGI-2 Technical Report.* arXiv:2603.06590 [cs.CL], 2026.
30. C. Talley, V. Tibrewal, S. Adekunle, W. Dong, X. Wu, F. Sheikh. *Multi-Perspective Transformers in ARC-AGI-2 Challenge.* arXiv:2605.01154 [cs.LG], 2026.
31. S. Iyengar. *Sample, Mix, or Verify? Allocating Test-Time Compute under Imperfect Verifiers.* arXiv:2610.23255 [cs.LG], 2026.
32. Q. Ma, Z. Tang, F. Meng, Z. Yao, H. Cui, W. Jia. *Correct but Unselectable: The Hidden Interface Tax in Multi-Candidate Reasoning.* NeurIPS 2026; arXiv:2610.32235.
33. M. Li, J. Qian, W. Mou. *What should post-training optimize? A test-time scaling law perspective.* arXiv:2605.10716 [cs.LG], 2026.
34. T. Zhou. *When RLVR Shrinks the Reasoning Boundary: Diagnosing Pass@k Inversion.* arXiv:2607.20543 [cs.LG], 2026.

35. L. Zhang, Y. He. *Test-Time Personalization: A Diagnostic Framework and Probabilistic Fix for Scaling Failures.* arXiv:2605.10991, 2026.
36. A. Pothula. *ARC-AGI-2: Why Test-Time Training Plateaus.* Kaggle dataset, CC0, 2026.
37. Z. Zhao et al. *Structured Sparse Memory for Recurrent Reasoning.* NeurIPS 2026; arXiv:2609.33270.
38. Nemhauser, G.L., Wolsey, L.A., Fisher, M.L. *An analysis of approximations for maximizing submodular set functions—I.* Mathematical Programming 14(1):265–294, 1978.
39. Markbarney et al. *Modality-Driven Search with Holistic Trace Judging for ARC-AGI-2.* 2026. https://arc.markbarney.net/paper.pdf
40. Y. Li, K.-S. Jun, K. Balagopalan, T. Nguyen, Y. Zhao, A. Zaidi, H. Nassif. *Halve-Then-Greedy: Adaptive Sampling for Posterior Success with Near-Optimal Instance Dependence.* arXiv:2610.20997, 2026.
41. Y. Huang, Y. Hou, V. Y. F. Tan. *Bayesian Fixed-Confidence Best-Arm Identification with Abstention.* arXiv:2610.30384, 2026.
42. Guancheng Pan, Qixin Tan, Yang Yuan, Andrew Chi-Chih Yao. *Algebraic World Model: From Exploration to Proofs.* arXiv:2610.40503 [cs.AI], 2026. https://ephemeral6.github.io/awm-site/

All web sources accessed 2026-10-18 (refs 26–30: 2026-10-20); arXiv identifiers resolve at `arxiv.org/abs/<id>`.

Appendix A — Pre-registration (condensed from `research/design/selection_protocol_v2.md`)

- **Sources frozen before evaluation:** B = `vanductan/arc-agi-2-candidate-dump` (pilot; 29 inputs/20 tasks — no other public candidate-dump notebook was found); A = CompressARC bundled predictions (claim-bearing; 400 ARC-AGI-1 eval tasks, fit on 400 train; hash-level candidates).
- **Reproduction gates:** B must reproduce 29 units, oracle 9, top-2 8; A must reproduce 137 solved and 80 pass@2 at 2,000 steps. A failed gate excludes the source; it does not trigger rule changes.
- **Units and splits:** B unit = test input, tasks split 9/11 dev/holdout by task hash (13/16 inputs); A unit = task, source-defined split (400/400).
- **Frozen scalar scores:** B = Σ of the eight per-augmentation scores (lower better); A = logaddexp-accumulated pick score (higher better).
- **Rules:** duplicate, argmax1_argmax2, top2_consensus, cal_indep, coverage_optimal, oracle_any2, deployed (selected on fit).
- **Features for $\hat{p}$ (fixed order):** score rank percentile, score z-score, log(1+agreement), cluster mass, is-top-cluster, shape consistency (B), cross-model count; A adds log first-step and pick-type diversity.
- **ρ classes:** same_candidate (ρ=1, never paired), same_view, same_source, cross_source (pooled only; no cross-source claim).
- **Statistics:** paired bootstrap over tasks, 2,000 resamples, seed 0; 80% and 95% CIs; claim iff $\Delta \geq +1.0$ point and 80% CI lower > 0 on the claim-bearing holdout.
- **Verdict:** Dataset A does not meet the pre-registered margin — coverage_optimal 17.75% vs argmax1_argmax2 19.50%, margin −1.75 pp (required: ≥+1.0 point with 80% CI lower bound >0) — so the selection claim is `DOWNGRADE_TO_ANALYSIS` and the deployed rule is the no-calibration fallback. Dataset C is simulated and therefore never enters the claim.
- **Deviations:** any change is a new protocol version committed before the affected holdout is read; both versions reported.

Appendix B — Evidence registry and per-number provenance

All numbers map to rows in `paper/evidence_table.md` (Plan C). Primary artifacts: `analysis/results/selection_benchmark.json` (Table 2 and Fig. 2; command `python analysis/selection_benchmark.py`); `logs/arch-theorist/external_reproduction_gate_arc2_dump.json` (Dataset B reproduction gate); `code/out/compressarc/verify_bundled_eval.json` (Dataset

A reproduction gate); `analysis/report.md` (data characterization); `analysis/results/saved_artifact_scores.json` (landscape); `logs/kaggle_freeze.md` and `logs/kaggle_runs.md` (platform gate); `code/docs/RESULTS.md` (CPU baselines and system validation). The theorem and the pre-registration are documented in `research/design/theory.md` and `research/design/selection_protocol_v2.md`, released with the paper bundle. GPU end-to-end execution and six scored leaderboard submissions are measured (Tables 8–9); no hidden-test data were accessed locally and no per-task hidden outcomes are available.

Appendix C — Platform-gated access log (summary)

Thirteen Kaggle runs recorded between 14:00 and 15:06 UTC on 2026-10-18, all CPU-only: five environment/mount/tokenizer probes, three fallback-packet dry runs, five TTT-packet smoke runs. GPU entitlement was requested (`NvidiaL4`, `NvidiaTeslaT4`) and not granted; competition attach was rejected pending rules acceptance, which requires phone verification. Zero Kaggle GPU-hours were consumed; the frozen handoff bundle contains packet checksums and a resume procedure that regenerates packets before pushing. No hidden-test data was accessed and no GPU number appears in the paper. **V4 update (2026-10-21):** the competition gate cleared; kernel v8 dev2 pilot measured on 4×L4 (§4.3). Submission **57456856** reran (2026-10-21T00:01:15Z) and errored before compute: the rerun branch set `ARC_BUDGET_S="41400.0"` and the run cell's `int("41400.0")` raised `ValueError`; Kaggle's ≈2 h 34 min was queue time. Hardening (int-typed budgets, no-raise validation, sanitization, SAFETY cell, rerun-simulation gate) landed; probe 57500870 scored 0.00; L4 refs 57663405=14.17, 57681632=16.25; S6 15.42.

Appendix D — Reproduction commands

```
# data analysis, split, baselines, allocation, figures (CPU, 169 s)
bash analysis/run_all.sh

# selection benchmark on third-party pools (CPU; needs the downloaded dumps)
python analysis/selection_benchmark.py

# repository tests and CPU baselines
python -m pytest code/tests -q
python code/compressarc_harness.py verify-bundled --out code/out/compressarc/
verify_bundled_eval.json
```

Appendix E — Diversification-threshold derivation, P1/P2 protocol, and falsifier

Theorem 4 and its corollaries are stated in §3.4; the full derivation, edge cases, and the exact inversion of the threshold are in `research/design/theory.md` §4.2. The empirical protocol is frozen in `research/design/selection_protocol_v2.md` §6d and implemented in `research/design/diversification_threshold.py` (self-test covers theorem arithmetic, admission/bootstrap mechanics and pilot flagging). On `pool-v1` logs from our own GPU runs: unit = (task, test input); task-level fit/holdout split by `sha256(task_id)` parity; $\hat{p}$ from the frozen dev-fit logistic calibration and $\hat{\rho}_c$ from the method-of-moments co-failure estimator per relation class (same_view, same_model_cross_view, score_tied, cross_model), clipped to $[-0.95, 0.99]$; $\delta=0.05$; pairs cap at the top 32 by $\hat{p}$. Rules compared on holdout units: top-1 duplicated, threshold-diversify (admit iff $\hat{\rho}_c < \rho^* - \delta$), and the ungated coverage-optimal contrast. Pre-registered predictions: **P1** admitted pairs beat their duplicate counterpart (paired $\Delta>0$, 80% CI>0); **P2** rejected (score-tied) pairs do not ($\Delta\leq 0$); **P3** predicted-vs-realized gain slope in $[0.7, 1.3]$. Falsifier: failure of P1, P2 or P3 falsifies the estimator/relation-class instantiation, never the algebra. The dev20 pools (65 base tasks / 133 task-pools) were analyzed; exact-match exclusivity makes P1/P2/P3 untestable (theory §4.8): every class has $\hat{\rho}<0$, so Table 10 is a prose check and Fig. 9 is omitted.

Appendix F — Official competition data note

The official evaluation files (`data/comp/`, sha256 prefixes `e7c62a4b…` / `84be4f4f…`) contain 120 tasks and **172** test inputs; the repo mirror distributed with the public data has 167. Five tasks differ: four add an extra test input (inserted first, so attempt order shifts), `faa9f03d` also differs in its train pairs, and `d8e07eb2` has a different train-pair output. Repo-mirror scores are therefore not official scores, and all official numbers in this paper are computed by `analysis/official_scoring.py --recompute-all`. The local `arc-agi_test_challenges.json` file (240 tasks / 259 inputs) is an exact content duplicate of 240 public training tasks and is a local dev sandbox only; it is never compared to the hidden leaderboard. Rerun budget: 4 workers × 12 h for ~240 tasks / 259 inputs gives 12.0 min per task / 11.1 min per input (10.8 / 10.0 min with a 10% margin). Dataset B's selection benchmark attached truth from the repo mirror; the one differing task present in B (`abc82100`) has no candidate matching any repository or official truth for its decoded input, so no rule covers it and the B row is unchanged (checked in `paper/official_b_truth_check.md`).

Appendix G — Theorem 5 derivation, dev20 protocol, and falsifiers

Appendix G / Table 11 (reserved): Theorem-5 derivation (Jensen, KKT/exchange, single-crossing, saturation); dev20 ladder S1/S2/S5/S5_ens ⟶ per-task wall, candidates, coverage, marginal gains by $\hat{p}$ proxies; PENDING/EXPLORATORY.

Appendix H — Exploratory cross-source note

An exploratory check on the partial saved artifacts (task-level 50/70 dev/holdout split, source quality estimated on dev only) found no conversion: on the 70-task holdout the best dev source solves 1/70, best-source attempt-1 plus cross-source attempt-2 also solves 1/70, while the union oracle is 8/70. Cross-source diversification does not convert the observed pool headroom on these partial artifacts. This is an appendix note only; it is not part of the pre-registered study, the artifacts are partial eval-mode runs rather than leaderboard reruns, and source quality was estimated on the dev split only.